3.3V DC–DC Buck converter

In 24V

+24V

470k/1% R53 · 75k/1% R54 · GND

U6 TP562933F: VIN, EN, BST, SW, SS, RT, GND, FB

C21 10uF/35V · C25 10uF/35V · C26 100nF · C27 100nF · C20 10nF · L1 10uH/4A · C22 22uF/10V · C23 22uF/10V · C24 100nF

3.28V feedback

R55 0 · R52 20k/1% · R11 62k/1%

3.3V out

+3.3V

input protection circuit

+24V

F1 Fuse_Small 4A · Q7 IRF4905 · J1 Conn_01x02_Socket · D2 RM3Z12 · D21 Green · R44 2.2k · GND

R43 100k

ok inducator

speed control input

+3.3V

J2 Conn_01x03_Socket

Speed ccontrol_ADC

GND

SENSING LOW PASS FILTERS

MCU_ADC_PH_A — R40 10 — 10nF C16 — GND — ADC_1

MCU_ADC_PH_B — R41 10 — 10nF C17 — GND — ADC_2

MCU_ADC_PH_C — R42 10 — 10nF C18 — GND — ADC_3

adc_1 adc_2 and adc_3 connecting op amp1,2and 3 Non-inverting pins

+3.3V

Speed control_ADC

STM32G431CBTx (U5)

VREF+, PG10, PF0, PF1, PC13, PC14, PC15, PB0, PB1, PB2, PB3, PB4, PB5, PB6, PB7, PB8, PB9, PB10, PB11, PB12, PB13, PB14, PB15, PA0, PA1, PA2, PA3, PA4, PA5, PA6, PA7, PA8, PA9, PA10, PA11, PA12, PA13, PA14, PA15, VSS, VSSA, VDD, VDDA

ADC_1 · ADC_2 · ADC_3

Ir2110 shut down_A · Ir2110 shut down_B · Ir2110 shut down_C

Hin_C · Hin_B · Hin_A · Lin_C · Lin_B · Lin_A

GND

IR2110 protect

Ir2110 shut down_A — R31 1k — Q8 BCV29 — R16 10k — D22 Red R45 220 — Shut down_A

Ir2110 shut down_B — R46 1k — Q9 BCV29 — R48 10k — D23 Red R47 220 — Shut down_B

Ir2110 shut down_C — R49 1k — Q10 BCV29 — R51 10k — D24 Red R50 220 — Shut down_C

+3.3V · GND

out motor drive phases

+12V · +24V · +3.3V · GND

PH_A

R7 1k 0.25W · D6 Green · D3 UF4007 · C1 4.7uF 24V · D10 Green · R9 2.2k

R8 330R 0.25W · D9 Green · D7 SR5100 · R2 10R · IRF540N Q1 · R10W1 · C2 4.7nF 50V · C5 470uF/50V

U1 IR2110S: VDD, VB, HO, HIN, LIN, SD, VSS, VS, VCC, LO · R1 100k

Hin_A · R13 1k · R12 10k · C6 22pF · Lin_A · Shut down_A

D8 SR5100 · R3 10R · IRF540N Q2 · R10W1 · C3 4.7nF 50V · R4 100k

MCU_ADC_PH_A — R14 10mR

PH_B

R17 1k 0.25W · D12 Green · D4 UF4007 · C7 4.7uF 24V · D15 Green · R26 2.2k

R10 330R 0.25W · D11 Green · D14 SR5100 · R20 10R · IRF540N Q3 · R10W1 · C8 4.7nF 50V · C12 470uF/50V

U2 IR2110S · R21 100k

Hin_B · R15 1k · R18 10k · C4 22pF · Lin_B · Shut down_B

D13 SR5100 · R19 10R · IRF540N Q4 · R10W1 · C9 4.7nF 50V

R22 100k

MCU_ADC_PH_B — R23 10mR

PH_C

R30 1k 0.25W · D17 Green · D5 UF4007 · C12 4.7uF 24V · D20 Green · R39 2.2k

R29 330R 0.25W · D16 Green · D19 SR5100 · R33 10R · IRF540N Q5 · R10W1 · C13 4.7nF 50V · C15 470uF/50V

U3 IR2110S · R34 100k

Hin_C · R27 1k · R28 10k · C11 22pF · Lin_C · Shut down_C

D18 SR5100 · R32 10R · IRF540N Q6 · R10W1 · C14 4.7nF 50V

R35 100k

MCU_ADC_PH_C — R36 10mR

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